

8DM60: Multimodal AI in Biomedicine

Integration of AI in the clinical workflow — Guided self-study

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How this session works

- Every case is a realistic deployment decision built on Lecture 5. Most have no single correct answer — the point is to reason through the trade-offs.
- Where numbers are given they are **hypothetical but plausible**.

Report back: 5-10 minutes per case

1. The **decision** you made
2. The **number** that drove it
3. The **risk** you could not mitigate
4. **Who** in the hospital owns that risk

Framework focus: **Q8** (how would it be used in practice?) and **Q9** (what could go wrong?).

The eight cases

	Case	Setting	Lecture 5 parts
1	A melanoma model meets dermatopathology	Pathology, research prototype	2 Scope and output, 6 Governance
2	One prostate model, three proposals	Pathology, CE-IVD product	3 Patterns, 6 Intended use
3	Choose the operating point	Pathology, lymph nodes	4 Metrics
4	The CE mark is not the end of the story	Pathology, PD-L1 IHC	4 Validation, 6 Intended use
5	Nothing is broken, everything is wrong	Pathology, HER2 IHC	4 Drift, 6 Governance
6	The model that cannot be proven wrong	Pathology, prostate IHC	4 Prediction paradox
7	Faster, cheaper, and nobody disagrees anymore	Radiology, chest X-ray	5 Human factors
8	Write the business case	Pathology, whole lab	7 Economics

Case 1 — A melanoma model meets dermatopathology

A university research group offers your lab its published model: **melanocytic lesion classification** (naevus vs. melanoma), AUROC 0.96 on 1 200 curated, balanced WSIs from one scanner.

It runs as a **web application** hosted by the university: you upload a slide, and after a few minutes you get a probability and a heatmap.

Your lab signs out **14 000 skin specimens per year**. The melanocytic excisions have on average 3 slides each; re-excisions of earlier melanomas are counted separately.

Skin specimen case mix (per year)

Basal cell carcinoma	30%
SCC and actinic keratosis	18%
Seborrhoeic keratosis, benign epidermal	14%
Inflammatory dermatoses	10%
Melanocytic lesions	**18%**
Re-excisions	7%
Other (cysts, adnexal, vascular)	3%

Your tasks

1. **Scope:** how many specimens per year does the model actually apply to? What happens to the rest, and who decides which slide is which?
2. **Unit of output:** the model gives one probability per slide. List what the melanoma report actually needs that the model does not give you.
3. **Is this allowed?** Name at least two legal or governance problems with using this tool in routine diagnostics as it stands.

Case 2 — One prostate model, three proposals

Your lab has licensed a **CE-IVD prostate biopsy cancer detection model**. It outputs $p(\text{cancer} \mid \text{slide})$ and a heatmap.

Its **intended use**: an adjunct to assist the pathologist, who reviews all slides of the case.

A typical biopsy case has **12 cores on 6 slides**. The lab has a backlog of three weeks, and three colleagues each make a proposal.

A — Worklist prioritisation. Rank each day's prostate cases by highest slide score, so suspicious cases are read first. Every case is still read.

B — In-viewer overlay. Show the heatmap as soon as a slide is opened, wherever $p > \tau$.

C — Benign fast track. When every slide of a case scores $p < 0.02$, the case is marked "AI benign" and signed out after a 30-second glance.

Your tasks

1. Which proposals are **inside the intended use**? What changes, legally and for your validation duties, if you implement the one that is not?
2. The model works per slide; the diagnosis is per patient. Propose an **aggregation rule** for A and for C (max, mean, any-positive, …). What does each choice do to case-level sensitivity?
3. Your head of department must choose one proposal tomorrow. Which one, and what is the single condition you would attach to it?

Case 3 — Choose the operating point

A vendor's model detects **breast cancer metastases in sentinel lymph node** slides (AUROC 0.97). Your lab reads **10 000 node slides per year** (250 working days); **5%** contain metastasis.

You must pick one of three operating points, for one of two uses:

(i) **Assistant:** the pathologist reads every slide; the model highlights suspicious regions.

(ii) **Rule-out:** model-negative slides are not reviewed.

	A	B	C
Sensitivity	0.99	0.95	0.85
Specificity	0.70	0.90	0.98
True positives			
False positives			
False negatives			
PPV			
NPV			
Flags per day			

Your tasks

1. **Fill in the table.** Show your working for one column.
2. Which operating point would you ship for use **(i) assistant**? And for **(ii) rule-out**? Justify each in one sentence about *whose* error matters more.
3. Under rule-out, how many metastases per year go **unseen by any human** at each operating point? How many slides per year does the pathologist no longer read?
4. The AUROC of 0.97 is identical for A, B and C. Explain in two sentences why it is nearly useless for this decision. **Who** in the hospital should sign off on the threshold?

Case 4 — The CE mark is not the end of the story

Your lab wants to buy a **CE-IVD model that scores PD-L1 tumour proportion score (TPS)** in non-small cell lung cancer. The TPS decides immunotherapy eligibility at the cut-offs **1%** and **50%**.

Vendor evidence: 3 centres, 600 resections and biopsies, intraclass correlation (ICC) of 0.90 against a panel of pathologists.

Pathologists are known to disagree most for cases *near* the cut-offs.

	Intended use	Your lab
Antibody	22C3	SP263
Stainer	Platform X	Platform Y
Scanner	Brand A	Brand B
Specimens	Histology	Histology *and* EBUS cytology cell blocks

Your tasks

1. Go through the table row by row. Which of your uses fall **inside** the intended use? What is your position — and your responsibility — for the ones that do not?
2. Why is an **ICC of 0.90** on the continuous score the wrong acceptance metric for this product? Propose a better one.
3. **Design the site-specific validation.** Specify: number of cases, how they are sampled (consecutive or enriched near the cut-offs?), the reference standard, the metric, and the acceptance criterion.
4. Pathologists disagree near the cut-offs. Then **what is ground truth**? How do you build a reference standard that is not just “whoever signed out”?

Lecture 5 guidance: validation should emulate the lab’s clinical environment, at least 60 routine cases per application (CAP/ASCP/API, 2022).

Case 5 — Nothing is broken, everything is wrong

A **HER2 IHC scoring model** (0 / 1+ / 2+ / 3+) went live in January as a concurrent assistant, validated with 0 and 1+ combined as “negative”.

What happened next

M4 You take over a neighbour’s breast surgery; post-neoadjuvant resections rise from 5% to 20%.

M6 A new therapy is approved for **HER2-low** tumours: 0 vs. 1+ now decides eligibility.

Period	0	1+	2+	3+	Override
Validation	25%	38%	22%	15%	—
M1–2	26%	37%	22%	15%	4.0%
M3–4	27%	36%	22%	15%	3.4%
M5–6	34%	31%	19%	16%	2.2%
M7–9	35%	30%	19%	16%	1.5%

Dashboard: share of cases per AI score; override = pathologist changed it.

No error was ever raised.

Your tasks

1. **M4:** the share of 0 rose after the takeover. Is it the **patients or the model**? What extra data would you pull to tell the two apart?
2. **M6:** the model was validated with 0 and 1+ combined. What is that validation **worth after M6**? Why does no dashboard number change on the day the therapy is approved?
3. **M6:** before M6, calling a 1+ tumour 0 cost the patient nothing. What **new validation** do you need now — which cases, which metric, which acceptance criterion?
4. **M4 + M6:** which patients are harmed by the combination? Estimate **roughly how many** per year if the lab scores 1 500 breast cancers annually.

Case 6 — The model that cannot be proven wrong

IHC ordering model

A pathology model predicts, for each prostate biopsy, whether **IHC is needed** to confirm the diagnosis. If it says “not needed”, no IHC is ordered.

After one year, an audit of the “not needed” cases finds **zero discrepancies** between the AI suggestion and the final report.

The vendor’s brochure now claims a **negative predictive value of 100%**.

Your tasks

1. Why is “zero discrepancies” **not** evidence of a 100% NPV? What was never measured?
2. **Design an evaluation** that can show whether the model is safe and helps patients. Consider randomisation (as in MASAI), held-out periods, IHC on a sample of “not needed” cases, and logging the model output and the action separately.
3. What are the **ethical and practical objections** to your design from task 2, and how would you answer them in front of the hospital’s ethics committee?

Case 7 — Faster, cheaper, and nobody disagrees anymore

A **chest X-ray detection model** (nodules, consolidation, pneumothorax) runs as a concurrent assistant. Boxes and a **"confidence 94%"** label appear on opening.

After six months

- Reporting time per study: −**22%**
- Radiologist–AI agreement: **91%** → **98.6%**
- A senior stopped using it after a visible early miss
- Two juniors felt unsure reading during an outage

Management now proposes

1. Raise the daily reporting target by **20%**.
2. **Disagreeing** with the AI requires a written justification in the report.
3. Report the override rate per radiologist monthly, aiming to **reduce** it.
4. Make the tool mandatory for all readers, including the senior.

Your tasks

1. Agreement rose to 98.6%. Does that show the AI is good, that radiologists have **calibrated** trust, or neither? What data would tell you which?
2. Which of the four management proposals are **reverse-centaur** symptoms? Rewrite the deployment policy so that the radiologist stays the centaur.
3. **Redesign the interface:** when is the output shown, how much is shown, how is it dismissed — and what replaces "confidence 94%"?
4. One radiologist overrides the AI with a justification and is **wrong**; another follows the AI and is wrong. Who is better protected legally, and how does that change behaviour?

Case 8 — Write the business case

A medium-sized Dutch lab (40 000 cases per year), **already fully digital**, is offered prostate biopsy cancer detection. Pathology is paid through a **bundled tariff**: the same fee per case with or without AI. Urology complains about turnaround time, and the lab has two pathologist vacancies. Figures are hypothetical.

Prostate biopsy cases per year	1 800	Cases currently needing IHC	30%
Licence, 3-year contract	€25 per case	Cost of IHC per case	€60
Integration, one-off	€40 000	Vendor claim: IHC reduction	40% (relative)
Pathologist cost	€150 per hour	Reading time per case	12 min
Validation + re-validation	0.1 FTE per year	Vendor claim: time saved	20%

Your tasks

1. Compute the **annual direct costs and benefits**. Does the tool pay for itself on IHC and reading time alone? State your assumptions.
2. Which benefits are **"fractional and distributed"**? Which would survive the finance director's question: *"which cost disappears?"*
3. Who **pays**, and who **captures** each benefit (lab, urology, patient, insurer)? What does the bundled tariff do to the argument?
4. If the efficiency case fails, is there still a case to buy? Make the strongest **non-financial** argument you can, and say what evidence it needs.

Closing discussion

1. In how many of the eight cases did the **AUROC** change your final decision?
2. Across all cases, **who** kept turning up as the owner of the risk you could not mitigate? Is that person usually in the room when the model is bought?
3. Pick one case and say which framework question (**Q1–Q9**) would have prevented the problem if it had been asked at the design stage.

Remember
A model that is never used has an AUROC of 1.0 and a clinical utility of zero.

And a model that is used without a validation plan, a monitoring plan and an exit plan has an unknown clinical utility — which is worse.